

# BAAZ & UDAY JKBOSE

2026 CLASS 11<sup>TH</sup>

## Units and Dimension

**LECTURE- 10** **PHYSICS**

**WAJAHID SIR (Gabbar Sir)**



# ***Topics To Covered***

**1**

**Short rivision**

**2**

**3**

**4**



Revision... ①

One Short

15 min

Unit and measurement

Gabber Sir



- Saare lecture dekh k Ahooo
- Copy samne Shagufta
- Full dekhna (15 min)



## Tips:-

- Apni Mehnat pa Barosa
- Apna App pa Yakeen
- Apna Sapno k Saat Fafadari

# Recap From 1st Lec.

Recap:-

Division

• Kasse parata hu

• Science

Physics

Chemistry

Biology

PO

$$PO = n u$$

• PO

Fundamental PO  
Basic PO

① ② ③ ④  
⑤ ⑥ ⑦

Derived PO

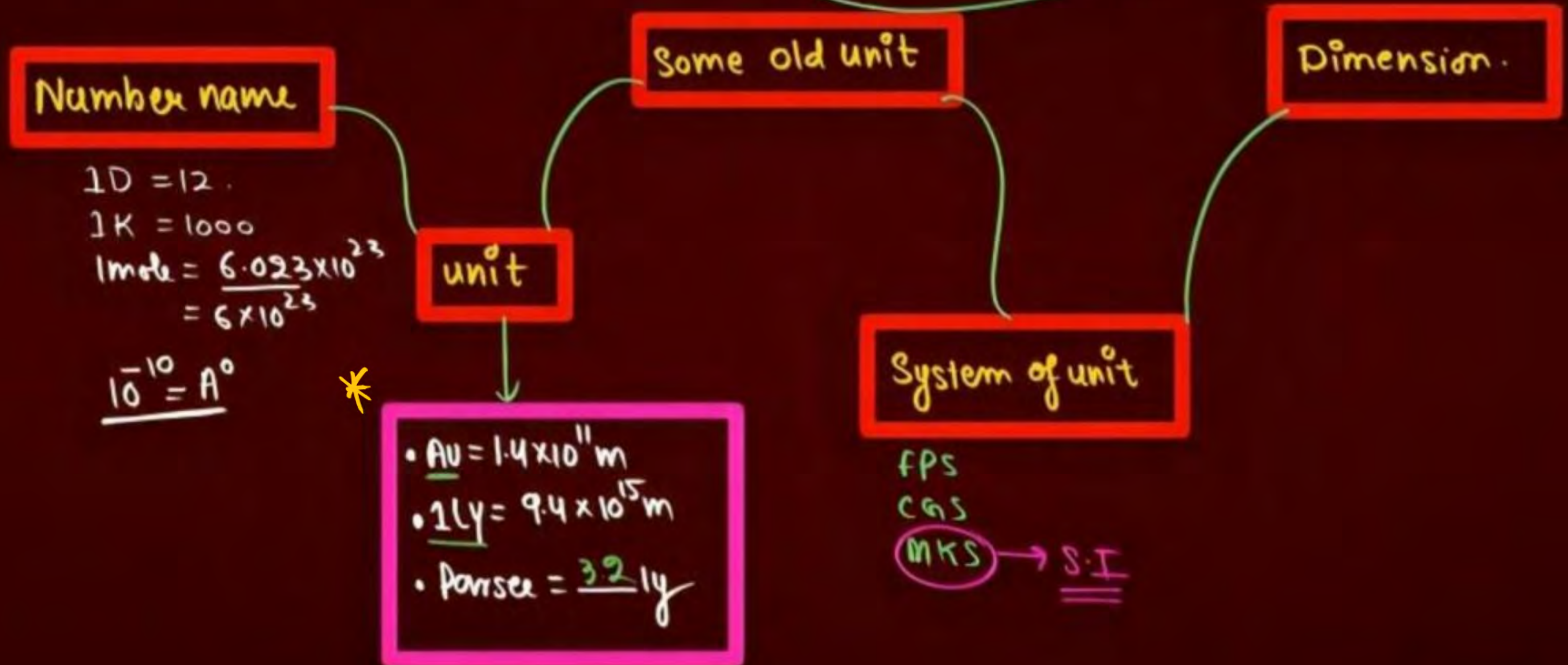
$$\text{Speed} = \frac{\text{distance}}{\text{time}} = \frac{m}{s}$$

\*\*  
Supplementary PO.  
SA

\* Number Name

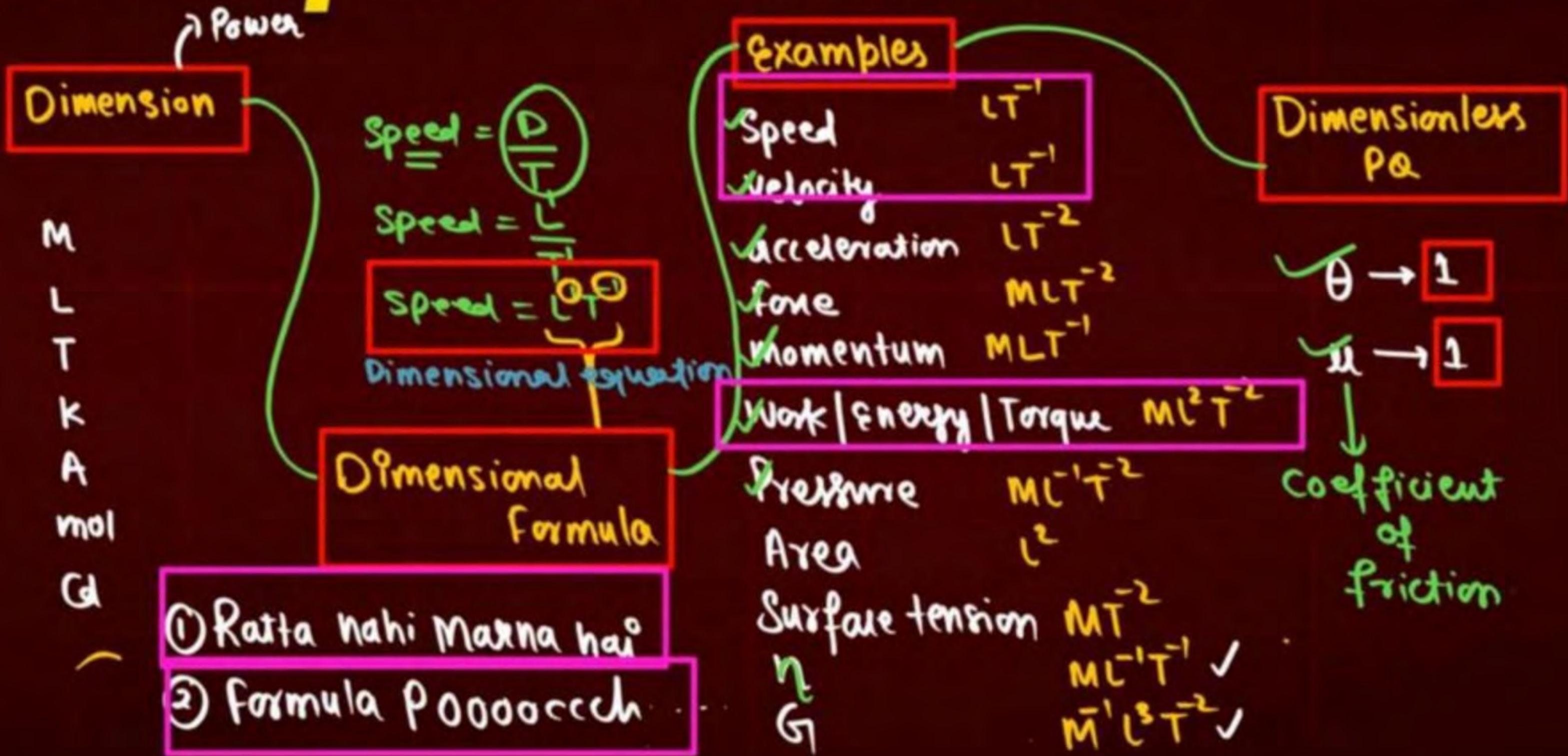


# Recap From 2. Lec.





# Recap From 3rd Lec.





# Recap From 4<sup>th</sup> Lec.

Application  
of  
Dimension

a ✓

b ✓

c

d

b To check the  
Correctness of  
Physics Formula

Law of  
Homogeneity

$$x + x = 2x$$

$$2x - x = x$$

a To find out the  
Dimensional  
Formula

$$\eta \rightarrow \text{ML}^{-1}\text{T}^{-1}$$

$$G \rightarrow \text{M}^{-1}\text{L}^3\text{T}^{-2}$$

①  $v = u + at$

$\downarrow$   
 $\text{LT}^{-1}$

②  $s = ut + \frac{1}{2}at^2$

$\downarrow$   
 $\text{L}$

③  $v^2 = u^2 + 2as$

$\downarrow$   
 $\text{L}^2\text{T}^{-2}$

④  $F = \frac{mv^2}{r}$

$\downarrow$   
 $\text{MLT}^{-2}$

$\downarrow$   
 $\text{MLT}^{-2}$



# Recap From 5<sup>th</sup> Lec.

Application of  
Dimension

© To derive the  
physics formula

Time period of Pendulum

• let  
 $T \rightarrow m^a l^b g^c$   
 $M^0 L^0 T^1 \rightarrow M^a L^b (LT^{-2})^c$   
 $a=0 \quad b=\frac{1}{2} \quad c=-\frac{1}{2}$



$2T = 2\pi \sqrt{l/g}$   
 $T = 2\pi \sqrt{l/g}$

Stroke law

• let  
 $F \rightarrow \eta^a r^b v^c$   
 $MLT^{-2} \rightarrow (ML^{-1}T^{-1})^a L^b (LT^{-1})^c$   
 $a=1 \quad b=1 \quad c=1$

$F = 6\pi \eta r v$



# Recap From 6<sup>th</sup> Lec.

④

conversion of  
one unit system  
into another  
unit system

$$* 1\text{m} = 100\text{cm}$$

$$1\text{kg} = 1000\text{gm}$$

$$1\text{N} = 10^5 \text{ dynes (PVA)}$$

$$1\text{joule} = 10^7 \text{ erg (PVA)}$$

$$n_2 = n_1 \left[ \frac{M_1}{M_2} \right]^a \left[ \frac{L_1}{L_2} \right]^b \left[ \frac{T_1}{T_2} \right]^c$$

\* Limitation of Dimension

\* we cannot derive the Dimensional constt.

\* we cannot derive the formula having  $(+/-)$

\* Cannot derive the formula with  $\sin\theta / \cos\theta$

\* It is not unique



# Recap From 7<sup>th</sup> Lec.

Imp

## Rounding off

## Significant Fig

Rule ①  $2.4829$   
 $2.483$

Rule ②  $2.4562$   
 $2.456$

Rule ③  $5.234215$   $5.234225$   
 $5.23422$   $5.23422$

Number

SF

24321

5

24000

2

34051

5

4.600

4

0.009

1

$3 \times 10^{-6}$

1

number

$\infty$  (infinite)



# Recap From 8<sup>th</sup> Lec.

## Scientific Notation

\* Ek tarika hai to  
reduce the number  $10^n$

$$245620. = 2.4562 \times 10^5$$

$$0.00052 = 5.2 \times 10^{-4}$$

## Why Measurement is important

- ✓ To understand the science
- ✓ Building Bridge & machines
- ✓ Cooking & Shopping
- ✓ Share clear information

**Thank**  
*You*